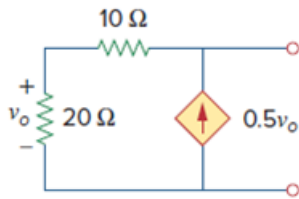


Problem

Find the Norton equivalent for the circuit in Fig. 4.129.

Figure. 4.129:



Step-by-step solution

Step 1 of 3

Refer to Figure 4.129 in the text book.

Connect a 1 V source across the output terminals to find the Norton resistance.

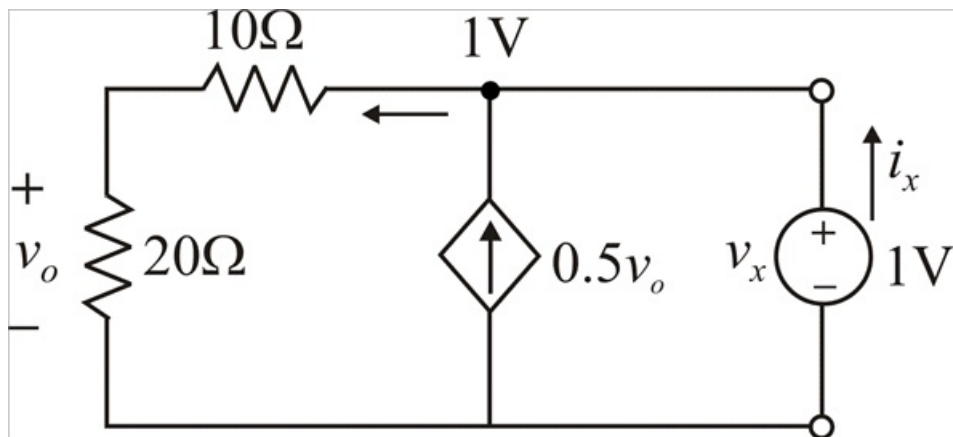


Figure 1

Step 2 of 3

Use voltage division principle to find the voltage v_o .

$$\begin{aligned} v_o &= 1 \text{ V} \cdot \frac{20}{20 + 10} \\ &= \frac{20}{30} \\ &= \frac{2}{3} \text{ V} \end{aligned}$$

Apply Kirchhoff's current law at node 1 V.

$$\begin{aligned} \frac{1}{10 + 20} &= 0.5v_o + i_x \\ \frac{1}{30} &= 0.5 \cdot \frac{2}{3} + i_x \\ \frac{1}{30} &= \frac{1}{3} + i_x \\ i_x &= \frac{1}{30} - \frac{1}{3} \end{aligned}$$

Calculate the Norton's resistance.

$$\begin{aligned} R_N &= \frac{1}{i_x} \\ &= \frac{1}{-0.3} \\ &= -3.333 \, \Omega \end{aligned}$$

Therefore, Norton equivalent resistance, R_N is $-3.333 \, \Omega$.

Since, the circuit does not have any independent source, the Norton's current is 0 A.

Thus, the Norton equivalent current, I_N for the given circuit is $0 \, \text{A}$.